Matrices & Matrix Algebra

ANSWER KEY

Section 1: Matrix Multiplication Properties

1 If matrix A has size 3×4 and matrix B has size 4×2, what is the size of the product AB?

a. 3×2 ✓

b. 4×4

c. 2×3

d. Multiplication is undefined

An $m \times k$ matrix multiplied by a $k \times n$ matrix produces an $m \times n$ matrix. Here $(3 \times 4) \times (4 \times 2) = 3 \times 2$.

2 Is matrix multiplication commutative in general ($AB = BA$)?

a. **No, $AB \neq BA$ in general even when both products are defined** ✓

b. Yes, multiplication of numbers and matrices is always commutative

c. Only if the entries are integers

d. Yes, provided both matrices are square

Matrix multiplication is fundamentally non-commutative: in general, $AB \neq BA$.

3 What is $(AB)^T$ equal to in terms of the transposes A^T and B^T?

a. $B^T A^T$ ✓

b. $A^T B^T$

c. AB^T

d. BA^T

The transpose of a product reverses order: $(AB)^T = B^T A^T$.

Section 2: Matrix Inverses and Elementary Matrices

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4 Find the inverse of the 2×2 matrix $A = [[4, 7], [1, 2]]$.

a. $[[2, -7], [-1, 4]]$ ✓

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

$\det(A) = 4(2) - 7(1) = 8 - 7 = 1$. The inverse is $\frac{1}{1}[[2, -7], [-1, 4]] = [[2, -7], [-1, 4]]$.

5 What is an elementary matrix?

a. **A matrix obtained by performing a single elementary row operation on an identity matrix** ✓

b. Any matrix containing only 0s and 1s

c. A diagonal matrix with determinant 1

d. A matrix with exactly one non-zero row

An elementary matrix E represents a single row operation; multiplying EA performs that row operation on A.

6 If A, B, C are invertible square matrices, what is $(ABC)^{-1}$?

a. $C^{-1}B^{-1}A^{-1}$ ✓

b. $A^{-1}B^{-1}C^{-1}$

c. CBA

d. ABC^{-1}

Inverting a product reverses the sequence of all factors: $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.

Section 3: Symmetric, Orthogonal, and Special Matrices

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7 A square matrix A is symmetric if and only if:

a. $A^T = A$ ✓

b. $A^T = -A$

c. $A^T = A^{-1}$

d. $\det(A) = 1$

A matrix is symmetric when it equals its own transpose: $A^T = A$ ($a_{ij} = a_{ji}$ for all i, j).

8 What defines an orthogonal matrix Q?

a. $Q^TQ = I$, **meaning** $Q^{-1} = Q^T$ ✓

b. $Q^2 = I$

c. $\det(Q) = 0$

d. All entries of Q are perpendicular

An orthogonal matrix has orthonormal columns, which means $Q^TQ = I$ and therefore $Q^{-1} = Q^T$.

9 If Q is an orthogonal matrix, what are the possible values for $\det(Q)$?

a. ± 1 ✓

b. 0

c. 1 only

d. Any real number

$\det(Q^TQ) = \det(I) = 1 \Rightarrow (\det Q)^2 = 1 \Rightarrow \det(Q) = \pm 1$.

Section 4: Linear Transformations and Standard Matrices

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10 A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear if and only if for all vectors $\mathbf{u}$, $\mathbf{v}$ and scalars c:

a. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ **and** $T(c\mathbf{u}) = cT(\mathbf{u})$ ✓

b. $T(\mathbf{u} \cdot \mathbf{v}) = T(\mathbf{u}) \cdot T(\mathbf{v})$

c. $T(\mathbf{u}) = \|\mathbf{u}\|$

d. $T(\mathbf{0}) = [1, 1, \ldots, 1]^T$

Linearity is defined by preservation of vector addition and scalar multiplication: $T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$.

11 What is the standard matrix for counterclockwise rotation in $\mathbb{R}^2$ by angle θ?

a. $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$ ✓

b. $[[\cos\theta, \sin\theta], [-\sin\theta, \cos\theta]]$

c. $[[\sin\theta, \cos\theta], [\cos\theta, -\sin\theta]]$

d. $[[\cos\theta, 0], [0, \sin\theta]]$

Rotating $[1, 0]^T$ gives $[\cos\theta, \sin\theta]^T$, and rotating $[0, 1]^T$ gives $[-\sin\theta, \cos\theta]^T$, giving $[[\cos\theta, -\sin\theta], [\sin\theta, \cos\theta]]$.

12 What is the standard matrix for reflection across the x-axis in $\mathbb{R}^2$?

a. $[[1, 0], [0, -1]]$ ✓

b. $[[-1, 0], [0, 1]]$

c. $[[0, 1], [1, 0]]$

d. $[[-1, 0], [0, -1]]$

Reflecting across the x-axis keeps x the same and negates y: $T([x, y]^T) = [x, -y]^T$, so the matrix is $[[1, 0], [0, -1]]$.

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